



Color image watermarking scheme based on singular value decomposition of split quaternion matrices

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ARTICLE INFO

Keywords:

Split quaternion matrices

SVDSQ

Color watermarking

Robustness

ABSTRACT

As digital media increasingly faces copyright threats, robust protection schemes are crucial. While gray-scale watermarking techniques are well-developed, color watermarking problems require more advanced solutions due to the complex spectral relationships among RGB channels. This paper introduces a novel color watermarking scheme based on the split quaternion matrix model, a 4D algebraic structure that preserves the linear spectral relationships among RGB channels, enhances computational efficiency, and improves color image processing. Watermark embedding employs a double split quaternion singular value decomposition (SVDSQ) process. The host image is divided into patches, and dominant singular values are extracted from each patch using SVDSQ. These values are assembled into a matrix, which undergoes a second SVDSQ to embed the watermark. This dual-layered method enhances the adaptability of the watermark embedding payload. Experimental results show that, under the same experimental conditions, our watermarking scheme demonstrates strong robustness against noise and geometric attacks, while maintaining a high peak signal-to-noise ratio (PSNR ≥ 35), compared to recent watermarking schemes. Despite introducing real part redundancy and requiring prior information for the watermark extraction, this approach advances color image processing using split quaternion matrices. Future work will focus on addressing these limitations and exploring new applications.

1. Introduction

With the widespread use of digital images, audio, and video, copyright protection has become increasingly important [1–4]. While gray-scale image watermarking techniques are relatively mature, color image processing still faces many challenges due to the complex spectral correlations among RGB channels. Color image watermarking techniques require not only embedding the watermark into the host image but also ensuring the robustness of the watermark extraction while maintaining image quality, which makes the implementation of these techniques more difficult. To address these issues, researchers introduced quaternions to represent color images, developing a quaternion matrix model to ensure color consistency and overall quality during image processing. However, while quaternions demonstrate excellent performance in processing color images, their limitations in computational complexity and reconstruction capability mean there is still room for further improvement in practical applications. To overcome these limitations, we introduce split quaternions (a variant of quaternions) into this field, defined as follows:

$$\mathbf{H}_s = \{h = h_1 + h_2i + h_3j + h_4k \mid h_1, h_2, h_3, h_4 \in \mathbf{R}\}, \quad (1.1)$$

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<https://doi.org/10.1016/j.jfranklin.2025.107508>

Received 29 April 2024; Received in revised form 8 October 2024; Accepted 1 January 2025

Available online 8 January 2025

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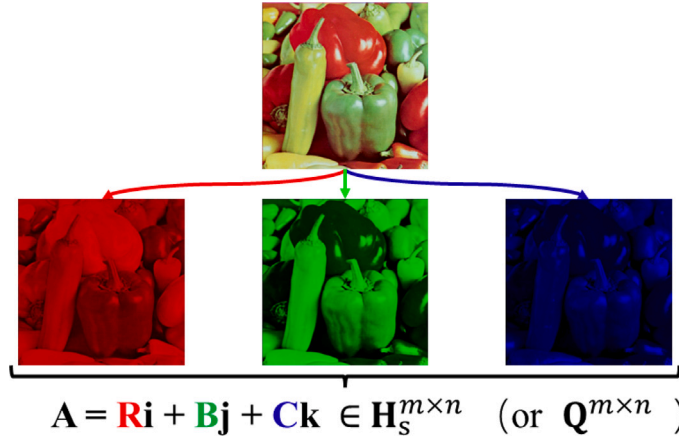


Fig. 1. Split quaternion (or quaternion) matrix model of color images.

where $\mathbf{H}_s$ denotes the set of split quaternions. h is a split quaternion, i, j and k are its three imaginary units, and their multiplication rule satisfy $i^2 = -1, j^2 = k^2 = 1, ij = -ji = k, jk = -kj = i, ki = -ik = j$. Color image model based on split quaternion matrices or quaternion matrices is shown in Fig. 1.

In the above model, each pixel of the color image “Peppers” is a combination of three primary colors: red (R), green (G), and blue (B) in a certain ratio. Each pixel can be considered a pure imaginary split quaternion (or a pure imaginary quaternion [5,6]). Therefore, the entire image can be represented by a pure imaginary split quaternion matrix (or a pure imaginary quaternion matrix). Although the model representations are the same in both algebras, there are significant differences in their computational rules and their ability to handle the corresponding color image problems, as shown below.

(1) Algebraic structures differ, meaning they have different operation rules. Multiplication rules for quaternionic imaginary units are $i^2 = j^2 = k^2 = -1, ij = -ji = k, jk = -kj = i$, and $ki = -ik = j$. $\mathbf{Q}$ denotes the set of quaternions. Additionally, SVDSQ exhibits characteristics that differ from the traditional SVD in the field of complex numbers and the ring of quaternions.

Lemma 1.1. (SVDSQ, [7]). Let $A \in \mathbf{H}_s^{m \times n}$. Then there exist two unitary split quaternion matrices $U \in \mathbf{H}_s^{m \times m}, V \in \mathbf{H}_s^{n \times n}$, such that

$$A = U \begin{bmatrix} \Sigma_r & 0 \\ 0 & 0 \end{bmatrix} V^H, \quad (1.2)$$

where $\Sigma_r = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_r), |\sigma_1| \geq |\sigma_2| \geq \dots \geq |\sigma_r| > 0$. $\sigma_1, \dots, \sigma_r$ are non-zero singular values of A and have a new form $\sigma = a + bj, a, b \in \mathbf{R}$.

(2) Algebraic isomorphisms are different. The most advanced known mappings for processing quaternion matrices and split quaternion matrices are as follows:

$$Q \xleftrightarrow{\phi} \begin{bmatrix} Q_1 & -\bar{Q}_2 \\ Q_2 & \bar{Q}_1 \end{bmatrix} \in \mathbf{C}^{2m \times 2n}, \quad H \xleftrightarrow{\varphi} \begin{bmatrix} H_1 + H_4 & H_2 + H_3 \\ -H_2 + H_3 & H_1 - H_4 \end{bmatrix} \in \mathbf{R}^{2m \times 2n}, \quad (1.3)$$

where $Q = H_1 + H_2i + H_3j + H_4k = (H_1 + H_2i) + k(H_4 + H_3i) = Q_1 + kQ_2 \in \mathbf{Q}^{m \times n}$ is a quaternion matrix, $Q_1 = H_1 + H_2i, Q_2 = H_4 + H_3i \in \mathbf{C}^{m \times n}$ are two complex matrices. $H = H_1 + H_2i + H_3j + H_4k \in \mathbf{H}_s^{m \times n}$ is a split quaternion matrix. $H_1, H_2, H_3, H_4 \in \mathbf{R}^{m \times n}$ are 4 real matrices.

Q is isomorphic to a structured matrix $\phi(Q) = \begin{bmatrix} Q_1 & -\bar{Q}_2 \\ Q_2 & \bar{Q}_1 \end{bmatrix}$ where the diagonal block matrices are conjugate

and the anti-diagonal block matrices are anti-conjugate. On the other hand, H is isomorphic to a non-structural matrix $\varphi(H) = \begin{bmatrix} H_1 + H_4 & H_2 + H_3 \\ -H_2 + H_3 & H_1 - H_4 \end{bmatrix}$, where $\phi()$ is a bijection. Obviously, H_1, H_2, H_3 and H_4 can all form quaternion matrices and split quaternion matrices of the same form, but due to the difference in algebraic structure, their mapping forms must satisfy the following equations

$$f(A + B) = f(A) + f(B), \quad f(AC) = f(A)f(C), \quad f(aA) = af(A), \quad (1.4)$$

where $f = \{\phi, \varphi\}$. A, B, C are three quaternion or split quaternion matrices satisfying the rules of addition and multiplication and a is a real number.

(3) Computational efficiencies are different. Due to the differences in the mapping forms of (1.3), the two algebraic matrix algorithms exhibit different computational efficiencies. Quaternion computations must preserve the structural form, while split quaternion computations maintain bijectivity at each step, eliminating the need to deliberately preserve the structure. For example,

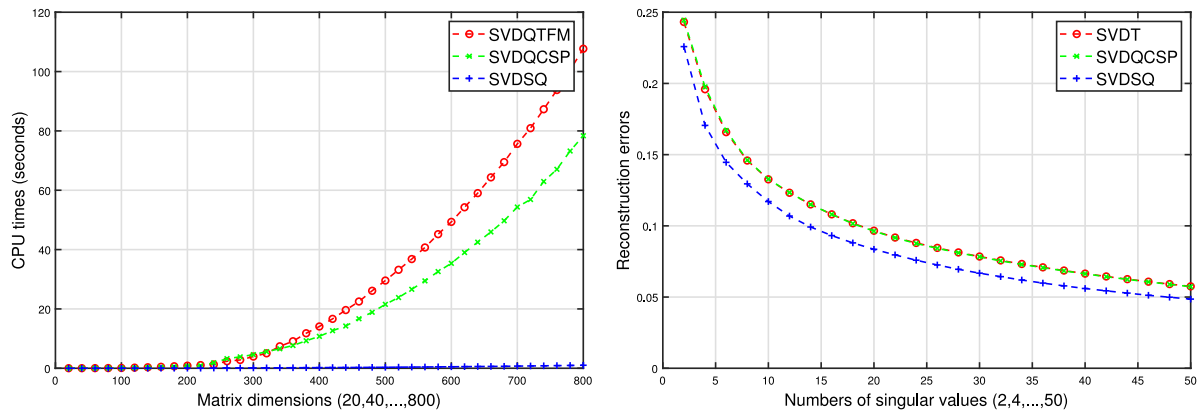


Fig. 2. Runtimes and reconstruction errors for several SVDs.

in the singular value decomposition of color images, the SVDSQ algorithm proposed in this paper offers advantages in both image decomposition speed and reconstruction performance.

The left side of Fig. 2 shows the CPU times required for decomposing simulated color images with resolutions of 20×20 , 40×40 , ... up to 800×800 pixels using quaternion SVD and SVDSQ. The simulated data settings are as shown in (1.5) and were executed with SVDQTFM, SVDQCSP and SVDSQ in MATLAB 2023b. $\text{round}()$ is the MATLAB function used for rounding to the nearest integer.

$$\begin{cases} m, n = 20, 40, \dots, 800; & R = \text{round}(255 \times \text{rand}(m, n)); \\ G = \text{round}(255 \times \text{rand}(m, n)); & B = \text{round}(255 \times \text{rand}(m, n)); \end{cases} \quad (1.5)$$

SVDQTFM represents the SVD algorithm from the Quaternion Toolbox [8], SVDQCSP refers to the structure-preserving algorithm proposed in [9,10], and SVDSQ corresponds to the algorithm presented in this paper. The right side of Fig. 2 shows the relative reconstruction errors when using three SVDs with a limited number of singular values to reconstruct the classic 512×512 pixel “Lenna” image. The reconstruction error is calculated as $\frac{\sqrt{\|R-\hat{R}\|^2 + \|G-\hat{G}\|^2 + \|B-\hat{B}\|^2}}{\sqrt{\|R\|^2 + \|G\|^2 + \|B\|^2}}$, where R, G, B represent the original image’s ($A = Ri + Gj + Bk$) three color channel matrices, and $\hat{R}, \hat{G}, \hat{B}$ represent the corresponding matrices of the reconstructed image. SVDT represents the error obtained by performing singular value decomposition and reconstruction separately on the R, G , and B . SVDSQ offers significant advantages, with singular values in the form of $a + bk$ and an isomorphic bijection with $2m \times 2n$ real matrix, resulting in efficient computation and excellent reconstruction quality. In contrast, traditional quaternion SVD (SVDQ) has singular values as real numbers and a structural mapping relationship with $2m \times 2n$ complex matrices, which leads to lower computational efficiency and reduced reconstruction accuracy. Therefore, SVDSQ excels in reducing real part redundancy and optimizing reconstruction performance. Additional theoretical algorithms and computational complexity comparisons related to quaternion matrices and split quaternion matrices can also be found in our previous works [11–18]. Real part redundancy is a common problem in the representation of 3D data using 4D algebras, leading to additional redundancy in the computation process. However, the split quaternion matrix model for color images can reduce this redundancy and make the computation more efficient. Therefore, this paper aims to develop a new color watermarking scheme by exploiting this advantage of the split quaternion algebra combined with the efficiency of SVDSQ. In addition, the improvements in color image processing algorithms based on this model and SVDSQ are not limited to the watermarking scheme proposed in this paper. They will also advance K-SVD dictionary learning, image feature extraction and recognition, and other related fields, which will be a focus of our future research.

In traditional image watermarking schemes, spatial domain schemes have low computational complexity but are generally less robust against image processing or geometric attacks. On the other hand, transform domain schemes, such as Discrete Cosine Transform (DCT), Discrete Wavelet Transform (DWT), and matrix decomposition schemes (e.g., QR, SVD decomposition), offer better robustness despite their higher computational complexity. Therefore, research into matrix decomposition schemes for color watermarking is crucial, as they enhance watermark robustness and effectively handle complex color data, thus improving practical application performance. Currently, several advancements have been made in color watermarking schemes based on matrix decompositions, including the following:

(1) Matrix decompositions-based blind watermarking schemes. In the paper [19], Su et al. leveraged the strong similarity correlation between the second-row first-column element and the third-row first-column element of the orthogonal matrix U after SVD. By modifying these elements, they embedded a binary watermark and used this relationship to extract the watermark as well. Following this work, Chen et al. extended this watermark embedding and extraction method to the SVD and QR decomposition of quaternion matrices. They improved the computational efficiency of the quaternion matrix decomposition by a fast structure-preserving algorithm to further enhance the anti-attack performance of the watermark extraction [20,21]. Additionally, in 2021, Sun et al. combined spatial domain techniques with singular value decomposition (SVD) methods to fully exploit the properties of maximum singular values, designing a blind color image watermarking scheme that embeds watermark information into the

maximum singular value (MSV) of a host image [22]. In the paper [23], Anh et al. proposed an image watermarking scheme based on Schur decomposition, which includes two different embedding algorithms. The first algorithm embeds the watermark into two components of the orthogonal matrix. In contrast, the second algorithm considers the embedding element and encodes the watermark image using the Arnold transform before embedding. Overall, binary color watermarking schemes are superior in resisting geometric attacks, effectively withstanding transformations such as rotation, scaling, and translation, and as blind watermarking schemes, facilitating watermark extraction and verification without the need for the original image. However, these schemes also have some obvious drawbacks. Since the watermarking information is embedded in the image through relationships between elements (e.g., numerical magnitude relationships, parity, and positivity), it is very sensitive to these subtle changes, making the scheme vulnerable to noise attacks. In addition, to embed watermark data (converted from decimal to binary form) into an image, it is usually necessary to maintain a large dimensional difference between the host image and the watermark—typically at least 12 times or more, which limits the embedding capacity of the watermark. Therefore, in practical applications, these advantages and disadvantages need to be weighed to select the appropriate scenario and technical parameters.

(2) Matrix decompositions-based non-blind watermarking schemes. In the paper [24], Li et al. proposed a color watermarking scheme based on real SVD and Arnold scrambling. The watermark image is scrambled using the Arnold transformation, and the watermark information is embedded by replacing the smallest singular values in the pixel blocks of the host image. Following this work, a similar color watermarking scheme based on the quaternion structure-preserving algorithm was proposed [9]. By utilizing the fast algorithm of SVDQ, this scheme significantly improves the invisibility of the watermarking scheme compared to the scheme in [24]. In the paper [25], Dappuri et al. proposed a non-blind color image watermarking scheme based on singular value decomposition in the translation-invariant wavelet domain, combined with an enhanced gray wolf optimizer to improve watermark extraction performance. In the paper [26], Patsariya et al. proposed a non-blind SVD-LWT watermarking scheme that utilizes the Y channel of the YCbCr color model for data embedding and combines a two-level LWT with an SVD algorithm. Such schemes generally embed watermarks into the host image through feature addition or replacement. Their advantages include a large embedding capacity, strong resistance to non-geometric attacks, and low time requirements for embedding and extraction. However, they also have weaknesses, such as poor resistance to geometric attacks, the potential for local highlight artifacts in the extracted watermark due to direct numerical alterations, and the need for substantial key storage. From the above discussion, it is evident that designing a watermarking scheme with robustness, large capacity, high security, and fast processing speed, based on the characteristics of matrix decompositions, is an urgent problem.

We design a novel color watermarking scheme that combines a double SVDSQ process, efficient feature preservation of singular values in the form of $a+bk$, and the effective retention of color information through 4D algebras. Compared to traditional quaternion matrix operations, this approach significantly reduces computational complexity. This scheme not only improves the adaptability of watermark embedding capacity but also enhances robustness. The new contributions of this work can be summarized as follows:

- This work extends the color watermarking problem to the domain of the split quaternion algebra for the first time, utilizing its algebraic structure and the new SVDSQ method to effectively enhance color information retention and computational speed.
- By partitioning the host image into patches and using double SVDSQ to embed dominant singular values, the approach increases embedding capacity and allows for adjustable embedding, thereby enhancing the adaptability of the watermarking scheme.
- The proposed scheme effectively improves the performance of the watermarking technique, with enhancements in both invisibility and robustness compared to some recent schemes.
- The improvements to color image processing algorithms based on the split quaternion matrix model and SVDSQ are not limited to the watermarking scheme proposed in this paper. They will also drive progress in related fields such as K-SVD dictionary learning, image denoising, image feature extraction, and recognition, paving the way for new research directions.

The paper is structured as follows. In Section 2, we mainly introduce the relevant theoretical knowledge to digital watermarking technology, including the basic definitions and algorithms of the split quaternion algebra, SVDSQ and of Arnold transform. In Section 3, we provide a detailed explanation of the specific implementation process for our watermarking scheme. In Section 4, we evaluate the performance of the proposed watermarking scheme from multiple perspectives, including watermark payload capacity, embedding and extraction time, invisibility, robustness, and storage overhead of keys, and compare it with several state-of-the-art watermarking schemes. In Section 5, we summarize the work presented in this paper and discuss potential directions for future research.

2. Preliminary mathematics

I. Split quaternion algebra. After the introduction of quaternions, split quaternions represent a further theoretical innovation in 4D algebraic systems, first proposed by J. Cockle in 1849 [27]. Quaternions and split quaternions are associative and non-commutative algebras, meaning they do not follow the commutative rule of multiplication. This non-commutativity has, to some extent, constrained their development. However, the development of mapping techniques has enabled the conversion of 4D algebraic problems into corresponding real or complex number problems, overcoming this limitation. As a result, quaternions and split quaternions have become a significant research focus in the field of algebras. The relevant symbols for $\mathbf{H}_s$ are listed in Table 1 below.

Table 1
Symbol table of split quaternions.

Symbols	Names	Descriptions
$\mathbf{R}$	the field of real numbers	
$\mathbf{H}_s$	the ring of split quaternions	
h	split quaternion	$h = h_1 + h_2i + h_3j + h_4k \in \mathbf{H}_s$
$ h $	modulus	$ h = \sqrt{ h\bar{h} } = \sqrt{ h_1 ^2 + h_2^2 - h_3^2 - h_4^2}$
A	split quaternion matrix	$A = A_1 + A_2i + A_3j + A_4k \in \mathbf{H}_s^{m \times n}$
$\bar{A}$	conjugate	$\bar{A} = A_1 - A_2i - A_3j - A_4k \in \mathbf{H}_s^{m \times n}$
A^T	transpose	$A^T = A_1^T + A_2^Ti + A_3^Tj + A_4^Tk \in \mathbf{H}_s^{n \times m}$
A^H	i-conjugate transpose	$A^H = A_1^T - A_2^Ti + A_3^Tj + A_4^Tk \in \mathbf{H}_s^{n \times m}$
U	unitary	$UU^H = U^H U = I_n, U \in \mathbf{H}_s^{n \times n}$

II. SVDSQ. The existence and non-uniqueness of SVDSQ were first proven and introduced in the authors' earlier work [7]. The SVDSQ theorem is provided in Lemma 1.1. In this work, we present a new construction and algorithm. Unlike previous forms of SVDSQ, this new approach more effectively preserves image information, as demonstrated in Fig. 2.

(1) SVD of the isomorphic real matrix $\varphi(A)$. Given $A = A_1 + A_2i + A_3j + A_4k \in \mathbf{H}_s^{m \times n}$. Then there exist two matrices $\mathbb{U} \in \mathbf{R}^{2m \times 2m}$ and $\mathbb{V} \in \mathbf{R}^{2n \times 2n}$ such that

$$\varphi(A) = \mathbb{U} \begin{bmatrix} \mathbb{D} & 0 \\ 0 & 0 \end{bmatrix} \mathbb{V}^H, \quad (2.1)$$

where $\varphi(A) = \begin{bmatrix} A_1 + A_4 & A_2 + A_3 \\ -A_2 + A_3 & A_1 - A_4 \end{bmatrix} \in \mathbf{R}^{2m \times 2n}$, $\mathbb{U} = [u_1, \dots, u_{2m}]$, $\mathbb{V} = [v_1, \dots, v_{2n}]$ are two unitary matrices, $u_g \in \mathbf{R}^{2m \times 1}$, $v_h \in \mathbf{R}^{2n \times 1}$, $g = 1, 2, \dots, 2m$, $h = 1, 2, \dots, 2n$ and $\mathbb{D} = \text{diag}(\tau_1, \tau_2, \dots, \tau_p)$, $\tau_1 \geq \tau_2 \geq \dots \geq \tau_p > 0$, $\tau_1, \tau_2, \dots, \tau_p$ are the non-zero singular values of $\varphi(A)$.

(2) Inverse construction of SVDSQ based on the SVD of $\varphi(A)$. According to Eq. (1.3), any split quaternion is isomorphic to a 2×2 real matrix. When a 2×2 diagonal matrix is formed by two adjacent singular values of $\mathbb{D}$, this diagonal matrix corresponds to a unique split quaternion, i.e.,

$$\begin{bmatrix} \tau_1 & 0 \\ 0 & \tau_2 \end{bmatrix} \xleftrightarrow{\varphi} \sigma_1 = \frac{\tau_1 + \tau_2}{2} + \frac{\tau_1 - \tau_2}{2}k, \quad (2.2)$$

this process establishes a unique correspondence between the 2×2 real matrix and a split quaternion, forming a bijection. Additionally, we can similarly determine the other singular values of A : $\sigma_t = \frac{\tau_{2t-1} + \tau_{2t}}{2} + \frac{\tau_{2t-1} - \tau_{2t}}{2}k$, $t = 1, 2, \dots, r$, $r = \begin{cases} \frac{p+1}{2}, & \text{if } p \text{ is odd,} \\ \frac{p}{2}, & \text{if } p \text{ is even,} \end{cases}$ $\tau_{p+1} = 0$, p is the number of non-zero singular values of $\varphi(A)$.

Similarly, we can construct the singular vectors of the corresponding split quaternion matrix A based on the adjacent left singular vectors and adjacent right singular vectors of $\varphi(A)$, that is,

$$[u_1, u_2] = \begin{bmatrix} \hat{u}_1 & \hat{u}_2 \\ \hat{u}_1 & \hat{u}_2 \end{bmatrix} \xleftrightarrow{\varphi} \mu_1 = \frac{\hat{u}_1 + \hat{u}_2}{2} + \frac{\hat{u}_2 - \hat{u}_1}{2}i + \frac{\hat{u}_2 + \hat{u}_1}{2}j + \frac{\hat{u}_1 - \hat{u}_2}{2}k, \quad (2.3)$$

where $\hat{u}_1 \in \mathbf{R}^{m \times 1}$ is the first row to the m th row of u_1 , $\hat{u}_1 \in \mathbf{R}^{m \times 1}$ is the $(m+1)$ -th row to the $2m$ -th row of u_1 . $\hat{u}_2 \in \mathbf{R}^{m \times 1}$ is the first row to the m th row of u_2 , $\hat{u}_2 \in \mathbf{R}^{m \times 1}$ is the $(m+1)$ -th row to the $2m$ -th row of u_2 . Additionally, we can similarly determine the other singular vectors of A :

$$\begin{cases} \mu_t = \frac{\hat{u}_{2t-1} + \hat{u}_{2t}}{2} + \frac{\hat{u}_{2t} - \hat{u}_{2t-1}}{2}i + \frac{\hat{u}_{2t} + \hat{u}_{2t-1}}{2}j + \frac{\hat{u}_{2t-1} - \hat{u}_{2t}}{2}k, \\ \nu_t = \frac{\hat{v}_{2t-1} + \hat{v}_{2t}}{2} + \frac{\hat{v}_{2t} - \hat{v}_{2t-1}}{2}i + \frac{\hat{v}_{2t} + \hat{v}_{2t-1}}{2}j + \frac{\hat{v}_{2t-1} - \hat{v}_{2t}}{2}k, \end{cases} \quad (2.4)$$

where $\hat{u}_{2t-1}, \hat{u}_{2t-1} \in \mathbf{R}^{m \times 1}$, $\begin{bmatrix} \hat{u}_{2t-1} \\ \hat{u}_{2t-1} \end{bmatrix} = u_{2t-1}$, $\varphi(\mu_t) = [u_{2t-1}, u_{2t}]$, $t = 1, 2, \dots, m$; $\hat{v}_{2j-1}, \hat{v}_{2j-1} \in \mathbf{R}^{n \times 1}$, $\begin{bmatrix} \hat{v}_{2j-1} \\ \hat{v}_{2j-1} \end{bmatrix} = v_{2j-1}$, $\varphi(\nu_j) = [v_{2j-1}, v_{2j}]$, $j = 1, 2, \dots, n$. Therefore, combine (2.2), (2.3) and (2.4), there exist diagonal split quaternion matrix $\Sigma_r = \text{diag}(\sigma_1, \dots, \sigma_r)$ and two unitary split quaternion matrices $U = [\mu_1, \dots, \mu_m] \in \mathbf{H}_s^{m \times m}$, $V = [\nu_1, \dots, \nu_n] \in \mathbf{H}_s^{n \times n}$ such that

$$\varphi(A) = \mathbb{U} \begin{bmatrix} \mathbb{D} & 0 \\ 0 & 0 \end{bmatrix} \mathbb{V}^H \Leftrightarrow A = U \begin{bmatrix} \Sigma_r & 0 \\ 0 & 0 \end{bmatrix} V^H, \quad (2.5)$$

where $\varphi(U)$, $\varphi(V)$ are the column permutation matrices of $\mathbb{U}$ and $\mathbb{V}$, respectively.

Proposition 2.1. Let $A \in \mathbf{H}_s^{m \times n}$, $m \geq n$. Then its SVDSQ has the following properties:

• The existence of split quaternion equivalence classes leads to the non-uniqueness of SVDSQ. Suppose that $\sigma = a + bk$. Then there exists a unitary split quaternion p satisfying $\sigma \sim \check{\sigma}$, i.e., $\sigma = p\check{\sigma}p^H$. Moreover, $A = \sum_{t=1}^r u_t p_t (p_t^H \sigma_t p_t) (v_t p_t)^H = \sum_{t=1}^r \check{u}_t \check{\sigma}_t \check{v}_t^H$. For example, in the paper [7], the singular value form of SVDSQ is $a + bj$, while in this paper it is $a + bk$.

Table 2
Singular values of SVDSQ.

	Construction forms	Modulus	Numbers of zero divisors
[7]	$\sigma_i = \frac{\tau_i + \tau_{n+i}}{2} + \frac{\tau_i - \tau_{n+i}}{2} j$	$ \sigma_i ^2 = \tau_i \tau_{n+i} $	$\begin{cases} N(0), & \text{if } N(0) \leq n; \\ 2n - N(0), & \text{if } N(0) > n; \end{cases}$
Proposed	$\sigma_i = \frac{\tau_{2i-1} + \tau_{2i}}{2} + \frac{\tau_{2i-1} - \tau_{2i}}{2} k$	$ \sigma_i ^2 = \tau_{2i-1} \tau_{2i} $	0 or 1

Table 3
Visual reconstruction and quality metrics of the image “Peppers” using different SVDSQ constructions.

Numbers of singular values	10	20	30	40	50	60	70
[7]							
PSNR	18.8557	21.8151	23.8496	25.3394	26.5731	27.6180	28.5094
SSIM	0.8445	0.9020	0.9273	0.9435	0.9552	0.9633	0.9692
Proposed							
PSNR	21.8099	25.3149	27.5594	29.1917	30.4363	31.4897	32.3707
SSIM	0.9018	0.9431	0.9628	0.9728	0.9790	0.9831	0.9859

• The diversity of singular value construction forms leads to the non-uniqueness of SVDSQ. In Table 2, $\tau_i, \tau_{n+i}, \tau_{2i-1}, \tau_{2i}$ are singular values of $\varphi(A)$ and $\tau_1 \geq \tau_2 \geq \dots \geq \tau_{2n}$. $N(0)$ denotes the number of zeros in $\tau_1, \dots, \tau_{2n}$. Clearly, $\frac{\tau_i + \tau_{n+i}}{2} + \frac{\tau_i - \tau_{n+i}}{2} j$ and $\frac{\tau_{2i-1} + \tau_{2i}}{2} + \frac{\tau_{2i-1} - \tau_{2i}}{2} k$ are not necessarily the same equivalence class. [7] has more zero divisor singular values than those proposed in this paper. (Zero divisors are a special property of the split quaternion algebra, where $0 \neq x, y \in \mathbf{H}_s$ but $xy = 0$, meaning x and y are two zero divisors in the ring of split quaternions.) And since it was experimentally found that SVDSQ is more effective at preserving feature information in color images under the constructed forms of (2.2), (2.3), and (2.4), we use only these forms in the following experiments. Table 3 shows the visual reconstruction of the color image “Peppers” using two different SVDSQ constructions with 10, 20, ..., and 70 selected singular values. The corresponding PSNR and SSIM values (formulas provided later) indicate that the SVDSQ method proposed in this work achieves better reconstruction performance in both visual and numerical terms compared to previous approaches.

III. Arnold scrambling. Arnold scrambling [28] is a two-dimensional discrete transformation that first applies a permutation along the x -axis, followed by a permutation along the y -axis. Finally, modular arithmetic is used to map the pixel positions back into the image’s valid range, a process akin to cutting and refilling the image, which thoroughly shuffles the pixel locations and makes the image content difficult to recognize. Additionally, this transformation is reversible, allowing the image to be restored to its original state using the same steps. As a result, Arnold scrambling is widely applicable in image encryption and information hiding. The rules for scrambling and descrambling are shown in (2.6).

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \bmod N, \quad \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \bmod N, \quad (2.6)$$

where a, b, c and d are any positive integers such that $ad - bc = \pm 1$. In our experiments, we set $a = 1, b = 1, c = 1$, and $d = 2$. $(x, y), (x_1, y_1)$ and N denote the original pixel point, the new pixel point after processing, and the size of the image, respectively.

3. Proposed watermarking scheme

I. Watermark embedding. Color watermark embedding process is detailed in Fig. 3, which includes a flowchart and a comprehensive step-by-step description of each stage of the procedure.

- Step 1. Preprocessing of the color host image. Separate the host image into RGB three channel images and construct the corresponding pure imaginary split quaternion matrix $A = Ri + Gj + Bk \in \mathbf{H}_s^{m_1 \times n_1}$.
- Step 2. Preprocessing of the color watermark image. Adjust the dimensions to divide the 3D color watermark image into R, G, and B components, ensuring that the host image dimensions are integer multiples of the watermark dimensions. To enhance the security of the watermark, apply Arnold transformation to each component, and then construct the corresponding pure imaginary split quaternion matrix $W = R_W i + G_W j + B_W k \in \mathbf{H}_s^{m_2 \times n_2}$.

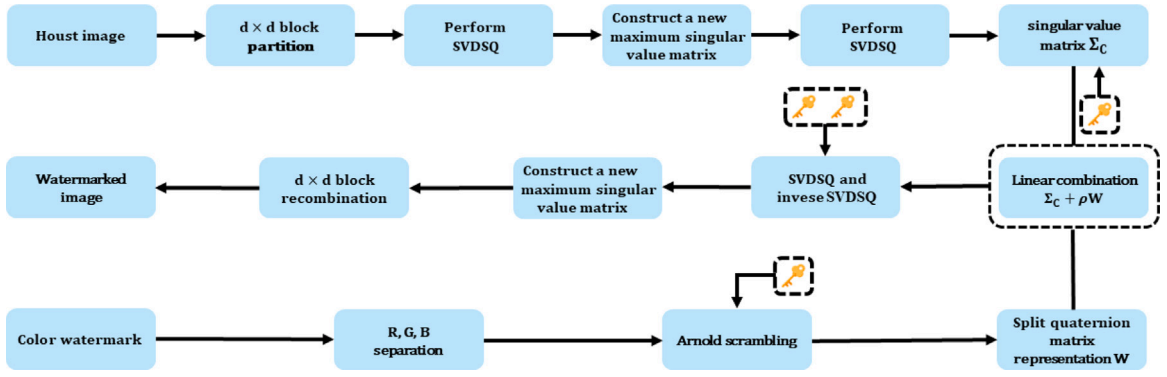


Fig. 3. Diagram of the watermark embedding process.

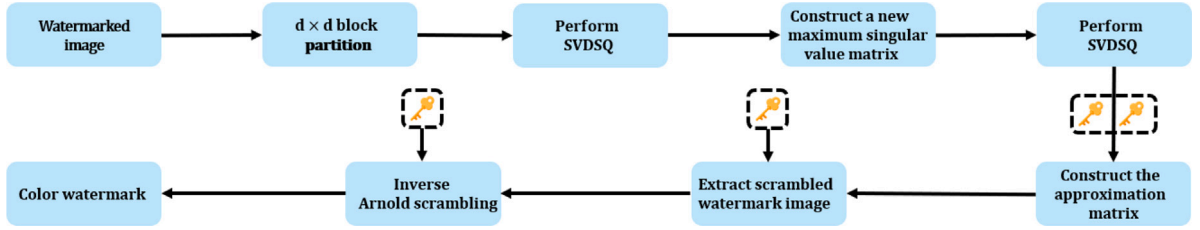


Fig. 4. Diagram of the watermark extraction process.

Step 3. Block processing and SVDSQ of A . To enhance the uniform invisibility of watermark embedding, divide the pure imaginary split quaternion matrix A into non-overlapping $d \times d$ patches ($d = \frac{m_1}{m_2}$), then perform SVDSQ on each patch.

$$A = \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1n_2} \\ A_{21} & A_{22} & \cdots & A_{2n_2} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m_21} & A_{m_22} & \cdots & A_{m_2n_2} \end{bmatrix} \xrightarrow{\text{SVDSQ}} \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1n_2} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2n_2} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{m_21} & \sigma_{m_22} & \cdots & \sigma_{m_2n_2} \end{bmatrix} = C,$$

where $A_{st} \in \mathbf{H}_s^{d \times d}$ is a patch of A , σ_{st} is the maximum singular value of A_{st} , $s = 1, 2, \dots, m_2$, $t = 1, 2, \dots, n_2$, $C \in \mathbf{H}_s^{m_2 \times n_2}$.

Step 4. Watermark embedding. The maximum singular values from all patches form a matrix C with the same dimensions as W . To enhance the robustness and reversibility of watermark extraction, the singular value matrix Σ_C of C is then linearly combined with W at a certain strength ρ , followed by performing SVDSQ:

$$C = U_C \Sigma_C V_C^H, \quad \Sigma_C + \rho W = U_{CW} \Sigma_{CW} V_{CW}^H.$$

Step 5. Obtaining the new maximum singular value matrix and the watermarked component image by performing inverse SVDSQ to all patches subsequently:

$$U_C \Sigma_{CW} V_C^H = \hat{C}, \quad U_{st} \begin{bmatrix} c_{st} & & & \\ & * & & \\ & & \ddots & \\ & & & * \end{bmatrix} V_{st}^H = \hat{A}_{st},$$

where c_{st} is the s th row and t th element of $\hat{C}$, $\hat{A}_{st}$ is the s th row and t th column watermarked patch.

Step 6. Recombine the watermarked patches $\hat{A}_{st}$ to obtain the watermarked image $\hat{A}$.

II. Watermark extraction. Color watermark extraction process is detailed in Fig. 4, which includes a flowchart and a comprehensive step-by-step description of each stage of the procedure.

Step 1. Preprocessing and block processing of the watermarked image. Separate the watermarked image into RGB three channel images and construct the corresponding pure imaginary split quaternion matrix $\hat{A} = \hat{R}i + \hat{G}j + \hat{B}k \in \mathbf{H}_s^{m_1 \times n_1}$.

Step 2. Block processing $\hat{A}$. Divide the pure imaginary split quaternion matrix $\hat{A}$ into non-overlapping $d \times d$ patches.

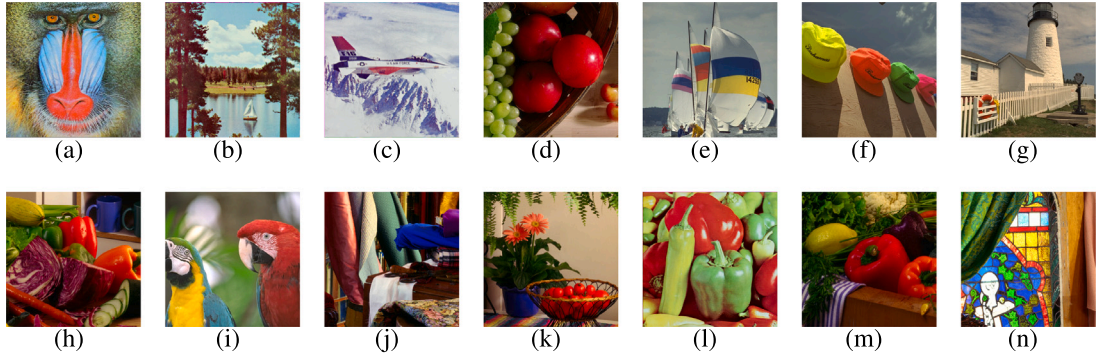


Fig. 5. All original color host images used in the experiments are named $H_a, H_b, \dots, H_n$ from left to right.

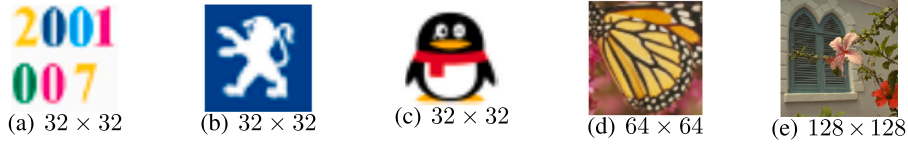


Fig. 6. All original watermark images used in the experiments are named $W_a, W_b, \dots, W_e$ from left to right.

Step 3. Double SVDSQ processing. Perform SVDSQ on each patch to construct the maximum singular value matrix $\hat{C}$, and then perform SVDSQ on $\hat{C}$,

$$\hat{A} = \begin{bmatrix} \hat{A}_{11} & \hat{A}_{12} & \dots & \hat{A}_{1n_2} \\ \hat{A}_{21} & \hat{A}_{22} & \dots & \hat{A}_{2n_2} \\ \vdots & \vdots & \ddots & \vdots \\ \hat{A}_{m_2 1} & \hat{A}_{m_2 2} & \dots & \hat{A}_{m_2 n_2} \end{bmatrix} \xrightarrow{\text{SVDSQ}} \begin{bmatrix} \hat{\sigma}_{11} & \hat{\sigma}_{12} & \dots & \hat{\sigma}_{1n_2} \\ \hat{\sigma}_{21} & \hat{\sigma}_{22} & \dots & \hat{\sigma}_{2n_2} \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\sigma}_{m_2 1} & \hat{\sigma}_{m_2 2} & \dots & \hat{\sigma}_{m_2 n_2} \end{bmatrix} = \hat{C},$$

$$\hat{C} = \hat{U}_C \hat{\Sigma}_C \hat{V}_C^H,$$

where $\hat{A}_{st} \in \mathbf{H}_s^{d \times d}$ is a patch of $\hat{A}$, $\hat{\sigma}_{st}$ is the maximum singular value of $\hat{A}_{st}$, $s = 1, 2, \dots, m_2$, $t = 1, 2, \dots, n_2$, $\hat{C} \in \mathbf{H}_s^{m_2 \times n_2}$.

Step 4. Watermark information extraction. Using the watermark extraction key and inverse SVDSQ to extract the watermark information.

$$U_{CW} \hat{\Sigma}_C V_{CW}^H = D, \quad \frac{D - \Sigma_C}{\rho} = \hat{W},$$

where U_{CW}, V_{CW} and Σ_C are prior information used for watermark extraction.

Step 5. Reconstruct watermark W by applying the inverse Arnold permutation to $\hat{W}$ using watermark extraction key.

4. Experimental analysis and comparison

I. Experimental setup and scheme. To ensure reproducibility, the experimental setup includes descriptions of hardware and software, experimental images, experimental schemes, and evaluation criteria.

(1) Experimental hardware and software. All experiments were performed and computed using MATLAB 2023b on a system equipped with a 13th Gen Intel(R) Core(TM) i7-13700K 3.40 GHz processor, 32 GB RAM, and Windows 11 Pro.

(2) Experimental images. Fig. 5 displays all the color host images used in the experiments, sourced from the CVG-UGR database [29] and the Kodak database [30], with each image resized to a resolution of 512×512 pixels. Fig. 6 shows all the color watermark images with different pixel resolutions used in the experiments.

(3) Analysis and comparison of watermarking schemes.

- Scheme 1: A binarization color watermarking scheme based on SVD of quaternion matrices [20].
- Scheme 2: A binarization color watermarking scheme based on SVD of real matrices [22].
- Scheme 3: A binarization color watermarking scheme based on QR decomposition of quaternion matrices [21].
- Scheme 4: A binarization color watermarking scheme based on Schur decomposition of real matrices [23].
- Scheme 5: A binarization color watermarking scheme based on Schur decomposition of real matrices [23].

Table 4

Payloads of several watermarking schemes.

	Scheme 1	Scheme 2	Scheme 3	Scheme 4	Scheme 5	Scheme 6	Scheme 7	Scheme 8	Scheme 9	Scheme 10
MP	≤ 45	≤ 45	≤ 45	≤ 45	≤ 45	≤ 128	= 512	= 512	≤ 128	≤ 512
MER	0.0618	0.0618	0.0618	0.0618	0.0618	0.0625	1	1	0.0625	1
AER	0.0312	0.0312	0.0312	0.0312	0.0312	0.0039	1	1	0.0039	0.0039

- Scheme 6: A non-binarization color watermarking scheme based on SVD of real matrices [24].
- Scheme 7: A non-binarization color watermarking scheme based on SVD of real matrices [25].
- Scheme 8: A non-binarization color watermarking scheme based on SVD of real matrices [26].
- Scheme 9: A non-binarization color watermarking scheme based on SVD of quaternion matrices [9].
- Scheme 10: Our color watermarking scheme.

(4) Evaluation criteria. In order to compare the visual fidelity of several watermarking schemes, the performance analysis metrics of these schemes are provided. These include the peak signal-to-noise ratio (PSNR), the structural similarity index (SSIM), and the normalized correlation (NC) between the original and processed images. Their formulas are defined as follows:

$$\text{PSNR} = 10 \lg \frac{3m^2 \times 255^2}{\sum_{x=1}^m \sum_{y=1}^m \sum_{z=1}^3 (H(x, y, z) - \hat{H}(x, y, z))^2},$$

where $H(x, y, z)$ represents the original host image, $\hat{H}(x, y, z)$ represents the watermarked image, (x, y) denotes the pixel coordinates, and $z = 1, 2, 3$ corresponds to the red, green and blue channels.

$$\text{SSIM} = \frac{(2\mu_H \mu_{\hat{H}} + C_1) (2\sigma_{H\hat{H}} + C_2)}{(\mu_H^2 + \mu_{\hat{H}}^2 + C_1) (\sigma_H^2 + \sigma_{\hat{H}}^2 + C_2)},$$

where H represents the original host image, $\hat{H}$ represents the watermarked image, $\mu_H, \mu_{\hat{H}}, \sigma_H, \sigma_{\hat{H}}$, and $\sigma_{H\hat{H}}$ are the local means, standard deviations, and cross-covariance for images H and $\hat{H}$, $C_1 = (0.01 \times L)^2$, $C_2 = (0.03 \times L)^2$ are the regularization constants for the luminance and contrast, L is the specified dynamic range value.

$$\text{NC} = \frac{\sum_{x=1}^n \sum_{y=1}^n \sum_{z=1}^3 W(x, y, z) \times \hat{W}(x, y, z)}{\sqrt{\sum_{x=1}^n \sum_{y=1}^n \sum_{z=1}^3 W(x, y, z)^2} \sqrt{\sum_{x=1}^n \sum_{y=1}^n \sum_{z=1}^3 \hat{W}(x, y, z)^2}},$$

where $W(x, y, k)$ represents the original watermark image, $\hat{W}(x, y, k)$ represents the restored watermark image, (x, y) denotes the pixel coordinates, $z = 1, 2, 3$ corresponds to the red, green and blue channels.

II. Analysis of watermark embedding payload. The importance of the watermark embedding payload lies in its flexibility. Both larger and smaller embedding payloads have their drawbacks: a larger payload may lead to a decline in image quality and increased visual loss. In contrast, a smaller payload may not fully utilize the data-hiding potential. Additionally, the size of the watermark carrier is sometimes uncontrollable. An adjustable embedding payload allows varying amounts of information to be embedded in the image, thereby enhancing data-hiding capability and the adaptability of the watermark. For a host image of $m \times m$ pixels, a watermark image of $n \times n$ pixels, and binarized watermarking schemes with a common 4×4 patch as an example, the formulas for the maximum payload (MP) and the embedding rate (ER) are as follows:

$$\text{MP} = \text{floor}\left(\sqrt{\frac{\frac{m}{4} \times \frac{m}{4}}{8}} = \frac{\sqrt{2}}{16} m\right), \quad \text{ER} = \frac{n \times n \times 8 \times 3}{m \times m \times 3},$$

where $\text{floor}()$ is the MATLAB function used to round down to the nearest integer. Specific experimental data ($m = 512, n = 512, 128, 64, 32$), MP, the maximum embedding rate (MER) and the actual embedding rate (AER) of all experimental schemes are shown in Table 4. Taking Scheme 1 as an example of the binarization watermarking scheme, a host image with a resolution of 512×512 pixels can accommodate a maximum watermark size of 45×45 pixels ($\text{floor}(\sqrt{\frac{512 \times 512 \times 3}{4 \times 4 \times 3 \times 8}}) = 45$), resulting in the maximum embedding rate $\text{MER} = \frac{45 \times 45 \times 8 \times 3}{512 \times 512 \times 3} \approx 0.0618$. In practical experiments, however, a watermark size of 32×32 pixels is typically used, yielding an actual embedding rate $\text{AER} = \frac{32 \times 32 \times 8 \times 3}{512 \times 512 \times 3} \approx 0.0312$. Taking Scheme 6 as an example of the non-binarization watermarking scheme, a host image with a resolution of 512×512 pixels can accommodate a maximum watermark size of 128×128 pixels ($\sqrt{\frac{512 \times 512 \times 3}{4 \times 4 \times 3}} = 128$), resulting in the maximum embedding rate $\text{MER} = \frac{128 \times 128 \times 3}{512 \times 512 \times 3} \approx 0.0625$. Additionally, since our watermarking scheme only requires that the watermark size does not exceed the size of the host image, the $\text{MP} \leq 512$, $\text{MER} = 1$.

III. Analysis of computational complexity and runtime. Matrix decomposition operations under different algebraic structures not only lead to significant variations in computational cost but also have a direct impact on the embedding and extraction speed of color watermarks. This impact is not only reflected in the difference in computational complexity but may also determine the efficiency and practicality of the watermarking process. Table 5 details the computational costs involved in various types of matrix (a color image of $m \times n$ pixels) decompositions across different watermarking schemes, providing a comparative analysis to help us better understand the performance of each scheme in practical applications. As shown in Table 5, Schur decomposition, and SVD based on

Table 5

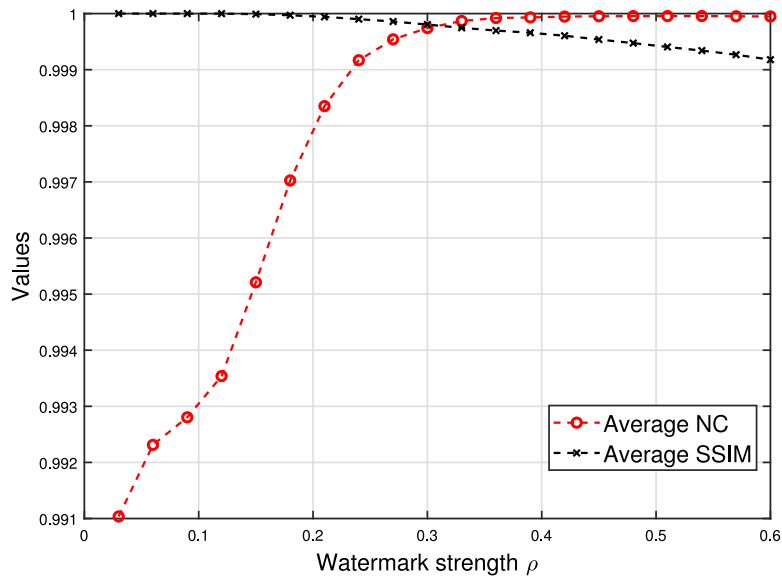
Computational cost of various matrix decompositions in experiments.

Schemes	Scheme 1	Scheme 2	Scheme 3	Scheme 4	Scheme 5	Scheme 6	Scheme 7	Scheme 8	Scheme 9	Scheme 10
Types	SVDQ	SVDR	QRQ	SchurR	SchurR	SVDR	SVDR	SVDR	SVDQ	SVDSQ
flops ($m \neq n$)	$64(mn^2 - \frac{n^3}{3})$	$3 \times mn^2$	$32(mn^2 - \frac{n^3}{3})$	$3 \times n^3$	$3 \times n^3$	$3mn^2$	$6 \times mn^2$	$2 \times mn^2$	$64(mn^2 - \frac{n^3}{3})$	$8mn^2$
flops ($m = n$)	$\frac{128}{3}n^3$	$3 \times n^3$	$\frac{64}{3}n^3$	$3 \times n^3$	$3 \times n^3$	$3n^3$	$6 \times n^3$	$2 \times n^3$	$\frac{128}{3}n^3$	$8n^3$

Table 6

Running times of several watermarking schemes.

Schemes(seconds)	Scheme 1	Scheme 2	Scheme 3	Scheme 4	Scheme 5	Scheme 6	Scheme 7	Scheme 8	Scheme 9	Scheme 10
Embedding	3.3906	0.6023	2.2872	0.3672	0.3783	0.1897	0.3836	0.1979	1.0893	0.3681
Extraction	2.9809	0.5117	1.9930	0.2863	0.2827	0.0111	0.2468	0.1318	0.0984	0.2946
Total	6.3714	1.1141	4.2802	0.6535	0.6611	0.2008	0.6304	0.3297	1.1876	0.6627

**Fig. 7.** Average values of SSIM and NC under different watermark strength $\rho = 0.03, 0.06, \dots, 0.6$.

R, G, B channels (or Y, Cb, Cr channels) have relatively lower computational complexity. Following these are the SVD based on split quaternion matrices, while QR decomposition and SVD based on quaternion matrices have higher computational complexity. These computational complexities only roughly reflect the watermark embedding efficiency from a mathematical perspective; different extraction methods will also impact the efficiency of watermark processing.

Host images (“Ha, Hb, ..., Hn”) and watermark images (“Wa, Wb, Wc”) were selected from their respective image sets and paired for the experiments. Table 6 presents the average time for embedding and extracting each watermarking scheme based on multiple trials. In Table 6, algorithm runtimes of less than 1 s are marked with colored numbers. It can be seen that the running time of the SVDSQ-based color watermarking scheme is nearly the same as that of the RGB three-channel watermarking schemes, indicating that the watermarking scheme using this 4D algebraic structure does not significantly increase the running time. However, the other quaternion watermarking algorithms have significantly longer runtimes. In addition, as Scheme 9 does not need to perform quaternion SVD again during the extraction process, the extraction time is faster, but the overall time is still slightly slower.

IV. Invisibility analysis. Given the significant impact of the intensity value ρ on the experiments, it is crucial to select an appropriate value using the evaluation above metrics. In the proposed scheme, the intensity value is determined through extensive experiments on standard 512×512 images from Fig. 7. As shown in Fig. 7, the average NC increases while the average SSIM decreases as the intensity value rises from 0.03 to 0.6. This indicates that as robustness improves, watermark imperceptibility diminishes. Therefore, considering the balance between imperceptibility and robustness, the intensity value ρ can be set to approximately 0.3 for the experiments to demonstrate the watermarking scheme’s balanced performance.

Table 7

Average PSNR and average SSIM of our scheme.

Watermark images	Wa	Wb	Wc	Wd	We
Average PSNR of watermarked imgs	47.5426	54.6010	54.0223	59.9173	54.8588
Average SSIM of watermarked imgs	0.9992	0.9998	0.9997	0.9999	0.9997
Average NC of extracted watermarks	1.0000	0.9999	0.9999	0.9995	0.9997

Table 8

Invisibility (PSNR/SSIM) comparison among different schemes.

Schemes	Scheme 1	Scheme 2	Scheme 3	Scheme 4	Scheme 5	Scheme 6	Scheme 7	Scheme 8	Scheme 9	Scheme 10
Average PSNR	33.0642	34.7067	36.6840	34.8653	37.2381	45.6181	37.8201	37.3832	46.7396	56.1884
Average SSIM	0.9255	0.9827	0.9699	0.9578	0.9673	0.9936	0.9887	0.9981	0.9952	0.9997
Average NC	0.9589	0.9983	0.9909	0.9959	0.9953	0.9995	0.9962	0.9971	0.9964	0.9998

Table 9

Various attacks used in experiments.

Index	Attacks	Description
A_1	Speckle noise corruption	Add speckle noise with variance $\sigma^2 = 0.02$ to the watermarked image
A_2	Poisson noise corruption	Add Poisson noise to the watermarked image
A_3	Gaussian noise corruption	Add Gaussian noise with variance $\sigma^2 = 0.01$ to the watermarked image
A_4	Salt & Peppers noise corruption	Add salt & peppers noise with variance $\sigma^2 = 0.05$ to the watermarked image
A_5	Darken	Adjust image intensity values from [0,1] to [0.4,0.8]
A_6	Brighten	Adjust image intensity values from [0,1] to [0.2,0.6]
A_7	Histogram equalization	Enhance the contrast of the watermarked image
A_8	Mean filtering	Apply average filtering with a 4×4 kernel to the watermarked image
A_9	Gaussian low-pass filtering	Apply Gaussian low-pass filtering with a standard deviation of 0.5 to the watermarked image
A_{10}	Rotation	Rotate the watermarked image 5° clockwise and then -5° counterclockwise
A_{11}	JPEG2000 compression	Adjust the watermarked image quality to 10% using JPEG2000 compression
A_{12}	Combination attack 1	$A_4 + A_{10}$
A_{13}	Combination attack 2	$A_{10} + A_{11}$

Table 7 demonstrates the results of embedding the watermarks “Wa”, “Wb”, ..., “Wc” from Fig. 6 into the original host image from Fig. 5 with an embedding intensity of 0.3, and then extracting them. For each watermarked image and each extracted watermark, the average PSNR is above 40, while both SSIM and NC are close to 1. These results indicate that, in the absence of attacks, the proposed watermarking scheme achieves high invisibility and excellent recovery performance.

Table 8 presents the average results of the experiments involving the pairwise combination of the original host images (“Ha, Hb, ..., Hn”) from Fig. 5 and the watermark images (“Wa, Wb, ..., Wc”) from Fig. 6, using the quantization values recommended in the original literature for embedding and extraction. It shows the average PSNR, SSIM, and NC for each scheme. Numerically, non-blind watermarking schemes demonstrate certain advantages due to their reliance on prior information for watermark extraction.

V. Robustness analysis. To fully assess the robustness of the watermarking scheme, it is crucial to evaluate its performance under various attacks. While the initial results show strong invisibility and recovery, real-world scenarios often involve attacks that could compromise watermark integrity. Therefore, testing the watermarking scheme under different attack conditions is essential for understanding its effectiveness in practical applications. Table 9 presents the experimental attacks, including noise addition (Speckle noise, Poisson noise, Gaussian noise, Salt & Peppers noise), brightness adjustment, contrast adjustment, filtering (Mean filtering, Gaussian low-pass filtering), rotation, JPEG compression, and combination attacks.

To compare the robustness of the color watermarking schemes as fairly as possible, Table 10 shows the extraction results using the same host image “Hc” and the watermark image “Wa”. The quantization parameters of each watermarking scheme are carefully selected to ensure that the PSNR of the watermarked image is greater than or equal to 35 as far as possible. The specific PSNR values of the 10 schemes are as follows: 35.0919, 35.6595, 35.0826, 35.0495, 35.0798, 42.6268, 35.4290, 35.5384, 40.9965, and 50.4103. The experimental design includes a no-attack test on the image after embedding the watermark, as well as the 13 attack tests listed in Table 9, after which the watermark is extracted. The robustness of these watermarking schemes was analyzed from multiple perspectives, demonstrating their resistance to interference under different attacks. From Table 10, it is evident that several binarization-based blind watermarking schemes exhibit poor robustness against noise and mean filter attacks. This is primarily due to their reliance on the binarization process, which makes the watermark susceptible to distortion caused by noise, leading to reduced accuracy during extraction. In contrast, non-binarized, non-blind watermarking schemes perform better against these attacks but lack robustness against geometric attacks. This weakness is mainly due to the difficulty in maintaining the stability of the watermark information during geometric transformations. The watermarking scheme proposed in this paper performs well under most of the listed attacks, mainly due to its low-dimensional embedding technique and double SVDSQ process. Specifically, we embed the

low-dimensional features of the watermark into the host image by modifying the maximum singular value of each image patch. Since the embedding operates in a low-dimensional space, it minimally affects the original host image. Additionally, unlike blind watermarking schemes, our scheme ensures precise extraction due to the mutually inverse nature of the embedding and extraction processes, which rely on prior information.

Similar to the experiments in Table 10, Tables 11 and 12 also evaluate the robustness of different watermarking schemes under varying conditions. In Table 11, “Hf” is used as the original host image and “Wb” as the watermark image, with the PSNR values of the watermarked images as follows: 35.3766, 34.6924, 36.3505, 35.4295, 35.0326, 45.0774, 35.3216, 38.3157, 43.4540, and 50.0650. In Table 12, “Hi” is used as the original host image and “Wc” as the watermark image, with the PSNR values of the watermarked images as follows: 31.3076, 34.4969, 35.3503, 35.2103, 35.0810, 43.7544, 35.5509, 41.9166, 41.9033, and 50.7538. These results demonstrate that our watermarking scheme continues to exhibit strong robustness across different experimental images.

Table 13 presents two comparative experiments focusing on non-blind watermarking schemes, which offer higher watermark capacity. Watermarks of 64×64 pixels “Wd” and 128×128 pixels “We” were selected, with “Hi” used as the host image. For these two sets of experiments, note that in Schemes 6 and 9, the watermark is embedded by replacing the smallest three singular values of each patch, so each patch must be at least 4×4 , allowing a maximum watermark size of 128×128 pixels in a 512×512 color host image. In contrast, Schemes 7 and 8 require that the watermark be embedded with the same dimensions as the host image, necessitating resizing the watermark to match the host image’s dimensions before embedding and restoring its original size after extraction. The PSNR values of the watermarked images, following the sequence of schemes listed in Table 13, are as follows: 45.3214, 36.6684, 35.0762, 41.7284, 50.8359, 33.8988, 35.7680, 37.5879, 36.5475, and 43.9503. These results demonstrate that our watermarking scheme continues to exhibit strong robustness, even with varying watermark sizes.

VI. Storage overhead for watermark extraction keys. The storage overhead for watermark extraction keys refers to the amount of storage required to keep the keys necessary for extracting watermarks from watermarked images. In blind watermarking schemes, where no prior information about the original image is used during extraction, the storage overhead for extraction keys is relatively small because these schemes typically rely on fixed or minimal information for key retrieval. In contrast, non-blind watermarking schemes, which use additional information from the original image during extraction, have a larger storage overhead for extraction keys. This is due to the need to store more detailed or complex keys that incorporate both the watermark and original image information. As a result, non-blind watermarking schemes generally require more storage resources compared to blind watermarking schemes.

For example, for a host image of size $m \times m$ pixels and a watermark of size $n \times n$ pixels, the storage overhead for extraction keys is shown in Table 14. It is evident that in blind watermarking schemes, the key consists only of the scrambling count and the sequence t for selecting patches, where $t \leq (\frac{m}{4})^2$, $m \geq n$. In Table 14, Scheme 1 requires only the scrambling count and the sequence t for selecting patches, resulting in a storage requirement of $1+t$. Scheme 2 requires the scrambling count, the quantization parameter, and the sequence t for selecting patches, leading to a storage requirement of $2+t$. Scheme 6 requires the scrambling count, the quantization parameter, and all orthogonal matrices U_{st} and V_{st} obtained from SVD on the patches, resulting in a storage requirement of $2+(3m)^2+m^2$, where $(3m)^2$ represents the storage overhead for all orthogonal matrices U_{st} and m^2 represents the storage overhead for all orthogonal matrix V_{st} . Scheme 10 requires the scrambling count, the quantization parameter, the orthogonal matrices U_{CW} , V_{CW} , and the diagonal elements of Σ_C , resulting in a storage requirement of $2+2 \times (2n)^2+2n$, where $2 \times (2n)^2$ represents the storage overhead for U_{CW} and V_{CW} , and $2n$ represents the storage overhead for Σ_C . All storage overheads are measured in decimal units, and the storage overhead for other schemes can be similarly calculated. Blind watermarking schemes typically rely on fewer key elements because they do not require additional data from the original image for watermark extraction. Consequently, the storage space needed for these keys is significantly smaller compared to that required by non-blind watermarking schemes. This advantage in key storage efficiency highlights the benefit of blind watermarking schemes, whereas non-blind watermarking schemes need to store more information to ensure accurate extraction, resulting in greater storage demands. This is also one of the limitations of the scheme proposed in this paper.

5. Conclusions

This paper presents a novel color watermarking scheme that leverages the split quaternion algebra and double SVDSQ to address the challenges of watermark embedding and extraction in color images with complex RGB spectral relationships. The innovative approach, which involves partitioning the host image into patches and embedding dominant singular values, effectively preserves color information, and enhances computational speed and the adaptability of watermark capacity. Compared to traditional methods, this scheme demonstrates strong resistance to most image processing attacks while maintaining a high peak signal-to-noise ratio (PSNR ≥ 35).

However, the scheme has limitations, including real part redundancy, dependence on prior information for extraction, and high key storage overhead. These factors may constrain its applicability in practical environments.

In future work, we will focus on optimizing this scheme to address these limitations and explore its performance across a broader range of images. Additionally, by leveraging the efficient characteristics of the split quaternion matrix model of color images and the SVDSQ approach, and integrating these with solutions to broader practical challenges, we aim to substantiate its effectiveness and extend its utility in the field of color image processing.

Table 10

Robustness results when embedding the “Wa” into “Hc” image.

	Scheme 1 $T=0.023$	Scheme 2 $T=46$	Scheme 3 $T=0.0313$	Scheme 4 $T=0.0335$	Scheme 5 $T=90$	Scheme 6 $T=0.16$	Scheme 7 $T=0.0075$	Scheme 8 $T=0.0175$	Scheme 9 $T=0.2$	Scheme 10 $T=0.25$
none										
NC	0.9753	1.0000	0.9992	0.9999	1.0000	1.0000	0.9985	0.9986	0.9965	1.0000
A_1										
NC	0.9238	0.8320	0.8985	0.9017	0.8904	0.8818	0.9662	0.9917	0.8869	0.9984
A_2										
NC	0.9610	0.8763	0.9426	0.9393	0.9707	0.9621	0.9877	0.9984	0.9572	0.9998
A_3										
NC	0.9237	0.8202	0.8993	0.9003	0.8851	0.8934	0.9696	0.9923	0.8946	0.9881
A_4										
NC	0.9426	0.9225	0.9889	0.9634	0.9020	0.8562	0.8760	0.7881	0.8629	0.9947
A_5										
NC	0.9282	0.7127	0.9866	0.9521	0.8962	0.8910	0.9564	0.9703	0.8637	0.9899
A_6										
NC	0.4394	0.3309	0.9604	0.9504	0.9258	0.5790	0.9424	0.9684	0.5439	0.9952
A_7										
NC	0.9691	0.7030	0.9970	0.9859	0.6729	0.9633	0.6549	0.3434	0.9582	0.9898
A_8										
NC	0.7745	0.9387	0.8621	0.8109	0.9480	0.4782	0.9557	0.9317	0.5338	0.9986
A_9										
NC	0.9673	0.9960	0.9932	0.9902	0.9983	0.9973	0.9938	0.9917	0.9955	0.9997
A_{10}										
NC	0.9394	0.9631	0.9819	0.9527	0.9688	0.9510	0.8787	0.7598	0.9469	0.9954
A_{11}										
NC	0.9718	0.9963	0.9960	0.9756	0.9962	0.8695	0.9982	0.9984	0.9048	0.9999
A_{12}										
NC	0.8726	0.8263	0.8637	0.8608	0.8571	0.6777	0.8604	0.7548	0.6705	0.9947
A_{13}										
NC	0.9336	0.9543	0.9794	0.9367	0.9662	0.8160	0.8793	0.7601	0.8339	0.9954

Table 11

Robustness results when embedding the “Wb” into “Hf” image.

	Scheme1 $T=0.04$	Scheme2 $T=46$	Scheme3 $T=0.05$	Scheme4 $T=0.06$	Scheme5 $T=90$	Scheme6 $T=0.16$	Scheme7 $T=0.0013$	Scheme8 $T=0.02$	Scheme9 $T=0.2$	Scheme10 $T=0.4$
none										
NC	0.9464	0.9922	0.9994	0.9931	0.9760	1.0000	0.9947	0.9945	0.9911	0.9999
A_1										
NC	0.9122	0.7767	0.8659	0.8606	0.8772	0.9072	0.9448	0.9510	0.9148	0.9960
A_2										
NC	0.9362	0.8562	0.9111	0.8784	0.9419	0.9484	0.9802	0.9857	0.9489	0.9982
A_3										
NC	0.8536	0.6975	0.8030	0.7990	0.7781	0.7949	0.9321	0.9600	0.8121	0.9934
A_4										
NC	0.8897	0.8405	0.9754	0.9172	0.7931	0.7739	0.8744	0.8622	0.7794	0.9913
A_5										
NC	0.7268	0.4930	0.8788	0.6918	0.5253	0.7512	0.7892	0.4859	0.7491	0.9748
A_6										
NC	0.8763	0.5863	0.9637	0.8239	0.6285	0.9459	0.8939	0.8736	0.9446	0.9685
A_7										
NC	0.9474	0.5862	0.9931	0.9250	0.6603	0.9511	0.8438	0.8409	0.9532	0.9779
A_8										
NC	0.6885	0.9051	0.7504	0.7006	0.8667	0.4514	0.9693	0.9316	0.4677	0.9972
A_9										
NC	0.9475	0.9879	0.9942	0.9770	0.9705	0.9950	0.9916	0.9867	0.9867	0.9995
A_{10}										
NC	0.9109	0.9493	0.9677	0.9133	0.9269	0.9344	0.9147	0.9041	0.9335	0.9950
A_{11}										
NC	0.9552	0.9912	0.9942	0.9354	0.9694	0.9090	0.9935	0.9928	0.9122	0.9997
A_{12}										
NC	0.7634	0.7156	0.7580	0.7600	0.7492	0.5984	0.8557	0.8352	0.5958	0.9818
A_{13}										
NC	0.9081	0.9458	0.9574	0.8881	0.9227	0.8265	0.9138	0.9051	0.8468	0.9950

Table 12
Robustness results when embedding the “We” into “Hi” image.

	Scheme1 $T=0.055$	Scheme2 $T=47$	Scheme3 $T=0.045$	Scheme4 $T=0.04$	Scheme5 $T=90$	Scheme6 $T=0.16$	Scheme7 $T=0.01$	Scheme8 $T=0.01$	Scheme9 $T=0.2$	Scheme10 $T=0.3$
none										
NC	0.9605	0.9972	0.9995	0.9991	0.9991	0.9976	0.9957	0.9948	0.9896	1.0000
A_1										
NC	0.9488	0.7938	0.8769	0.8612	0.9247	0.9067	0.8299	0.7207	0.9142	0.9951
A_2										
NC	0.9584	0.8581	0.9283	0.8955	0.9774	0.9595	0.9573	0.9734	0.9558	0.9976
A_3										
NC	0.9150	0.6986	0.8219	0.8139	0.7945	0.8360	0.8591	0.8813	0.8529	0.9931
A_4										
NC	0.9025	0.8534	0.9809	0.9412	0.8455	0.7933	0.7761	0.6717	0.8233	0.9887
A_5										
NC	0.7070	0.5425	0.8902	0.8117	0.6575	0.8723	0.8072	0.5708	0.8543	0.9864
A_6										
NC	0.8609	0.6705	0.9399	0.9268	0.7153	0.8471	0.8721	0.8924	0.8444	0.9896
A_7										
NC	0.9623	0.6284	0.9976	0.9857	0.7049	0.9668	0.8588	0.8416	0.9583	0.9971
A_8										
NC	0.6506	0.8810	0.7444	0.6727	0.9083	0.5240	0.9557	0.8011	0.5237	0.9942
A_9										
NC	0.9560	0.9918	0.9965	0.9816	0.9948	0.9927	0.9897	0.9576	0.9872	0.9992
A_{10}										
NC	0.9256	0.9529	0.9749	0.9313	0.9592	0.9359	0.7537	0.5917	0.9308	0.9921
A_{11}										
NC	0.9570	0.9953	0.9956	0.9632	0.9964	0.9529	0.9942	0.9927	0.9360	0.9998
A_{12}										
NC	0.8415	0.6965	0.7673	0.7700	0.7522	0.6171	0.7161	0.5672	0.6186	0.9890
A_{13}										
NC	0.9282	0.9503	0.9712	0.9044	0.9579	0.8861	0.7504	0.5935	0.8771	0.9920

Table 13

Robustness results when embedding the “Wd” and “We” into “Hl” image.

	Scheme6 $T=0.16$	Scheme7 $T=0.012$	Scheme8 $T=0.03$	Scheme9 $T=0.2$	Scheme10 $T=0.3$	Scheme6 $T=0.15$	Scheme7 $T=0.015$	Scheme8 $T=0.025$	Scheme9 $T=0.18$	Scheme10 $T=0.3$
none										
NC										
A_1										
NC										
A_2										
NC										
A_3										
NC										
A_4										
NC										
A_5										
NC										
A_6										
NC										
A_7										
NC										
A_8										
NC										
A_9										
NC										
A_{10}										
NC										
A_{11}										
NC										
A_{12}										
NC										
A_{13}										
NC	0.7935	0.6923	0.7859	0.8315	0.9935	0.6197	0.7221	0.5640	0.6800	0.9971

Table 14

Storage overhead for watermark extraction keys.

Schemes	Scheme 1	Scheme 2	Scheme 3	Scheme 4	Scheme 5	Scheme 6	Scheme 7	Scheme 8	Scheme 9	Scheme 10
Storage	$1 + t$	$2 + t$	$1 + t$	$1 + t$	$2 + t$	$10m^2 + 2$	$15n^2 + 3n + 2$	$10n^2 + n + 2$	$8m^2 + 2$	$8n^2 + 2n + 2$

CRedit authorship contribution statement

Gang Wang: Writing – original draft, Software, Methodology. **Tongsong Jiang:** Writing – review & editing, Project administration, Conceptualization. **Dong Zhang:** Validation, Resources, Investigation. **V.I. Vasil'ev:** Writing – review & editing, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The work of T. Jiang, V. I. Vasil'ev and D. Zhang is supported by the Russian Science Foundation No. 23-71-30013 (<https://rscf.ru/project/23-71-30013/>, Sections 1-3). The work of G. Wang is supported by the Russian Science Foundation No. 23-41-00037 and the Chinese Government Scholarship, China No. 202008370340 (<https://rscf.ru/project/23-41-00037/>, Section 4). The work of D. Zhang is supported by the Chinese Government Scholarship, China No. 202108370086 (Section 5). The authors would like to thank the anonymous referees for their valuable comments and suggestions, which led to substantial improvements in this paper. We also extend our gratitude to the editors for their support and guidance throughout the review process.

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